

[illegible]



netomat.net memory:17199 speed:343

netomat.net memory:17199 speed:343

netomat.net memory:17199 speed:343

netomat.net memory:17199 speed:343

network error. make sure you are connected and restart netomat.



netomat



NETWORK ERROR

netomat.net memory:17199 speed:343

netomat.net memory:17199 speed:343

netomat.net memory:17199 speed:343

netomat.net memory:17199 speed:343

network error. make sure you are connected and restart netomat.

netomat.net memory:14464 speed:333

txt:0 img:0 snd:0

network error. make sure you are connected and restart netomat.

netomat.net memory:17199 speed:343

netomat.net memory:17199 speed:343

netomat.net memory:17199 speed:343

netomat.net memory:17199 speed:343

network error. make sure you are connected and restart netomat.

Yannick Westphal

- > media artist
- > hacker
- > researcher

<https://hacking.art>

mail@hacking.art

- # Bachelor "Digital Media" University of the Arts Bremen
- # Diploma "Media Art" Academy of Media Arts Cologne

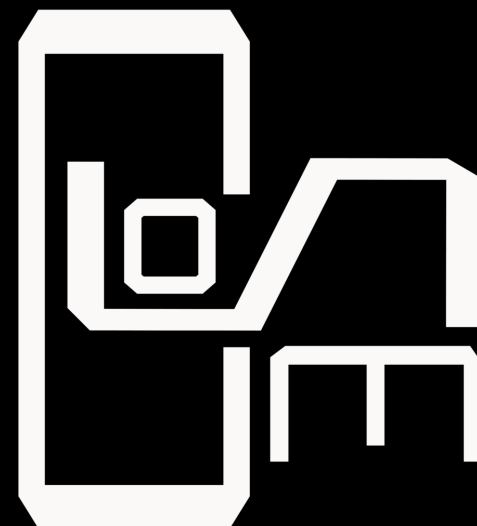
- # Almost 10 years of
working experience in the field of **IT security**

- # OSCP + OSWE + RTO certifications

- # published various articles & research on the subject

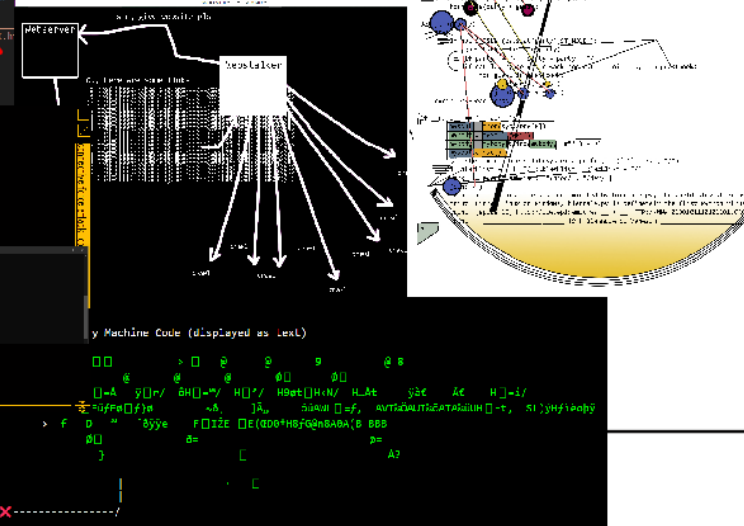
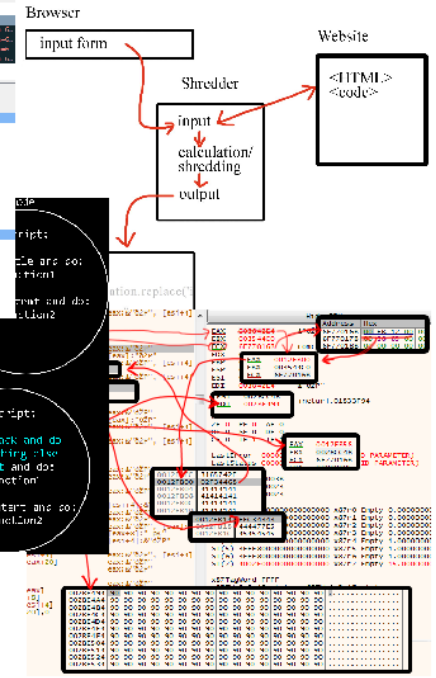
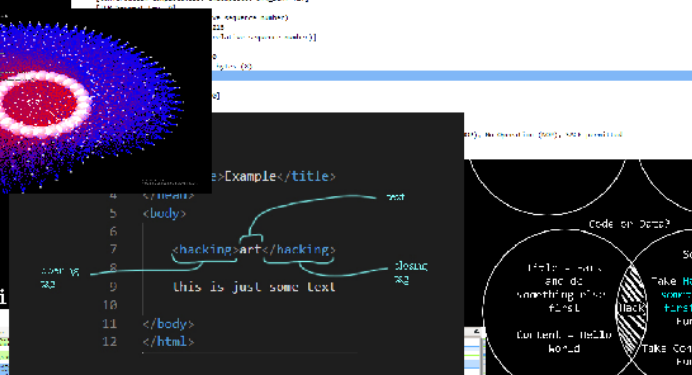
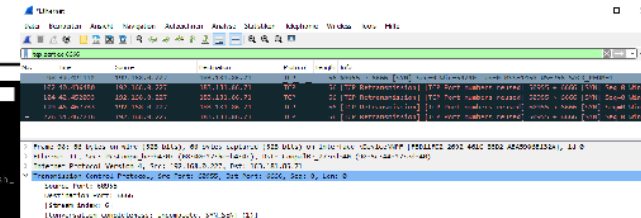
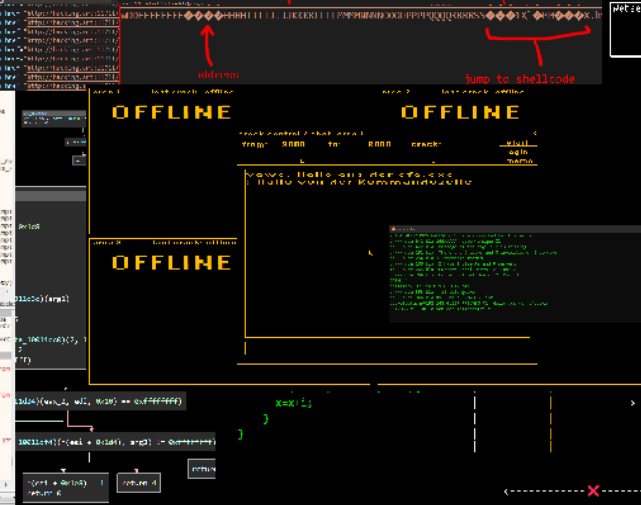
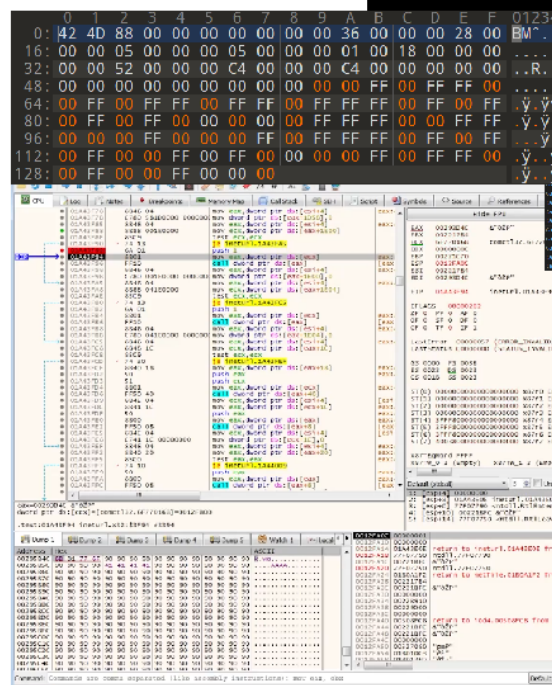
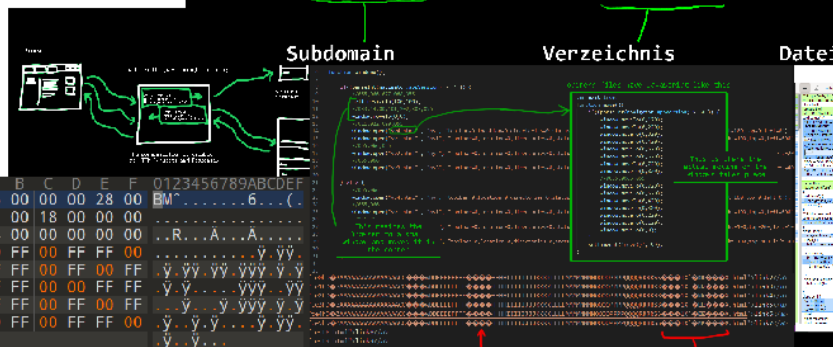
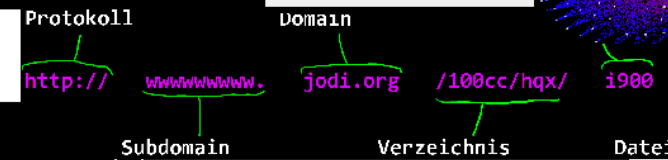
- Involved with **COSE** for 4 years

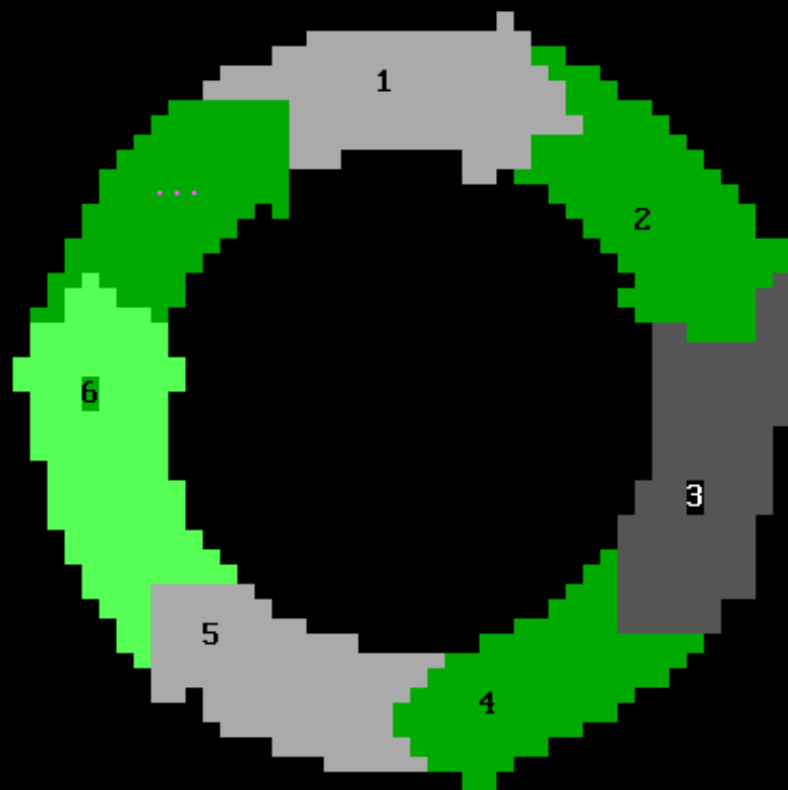
check out my work at hacking.art!



CODED SECRETS RESEARCH GROUP

Handy stuff





Reconnaissance

>> Collecting information

Scanning

>> Enumeration for attack vectors

Vulnerability Analysis

Exploitation

>> Inject malicious code

Post-Exploitation

>> Privilege Escalation

>> Lateral Movement

Reporting

>> Documentation

Remediation / Retest

Deconstruction

dismantle existing systems to reveal their logic, weaknesses, and hidden assumptions

Recombination

New forms emerge by rearranging fragments, whether through collage, code modification, or creative reuse

Recontextualization

Objects or technologies gain new meaning when placed in unfamiliar contexts

Subversion

Both practices deliberately challenge rules, norms, and established systems instead of simply accepting them

Provocation

Shock, irritation, or absurd interventions are used to attract attention and expose underlying tensions

Appropriation

Existing materials are not just reused, but examined, claimed, and transformed

Intermediality

Working across media, combining technology, language, images, objects, and performance

Transformation

The goal is not mere destruction, but the creation of new functions, meanings, and possibilities



Hacking Art: Software Analysis as Artistic Practice

Yannick Westphal

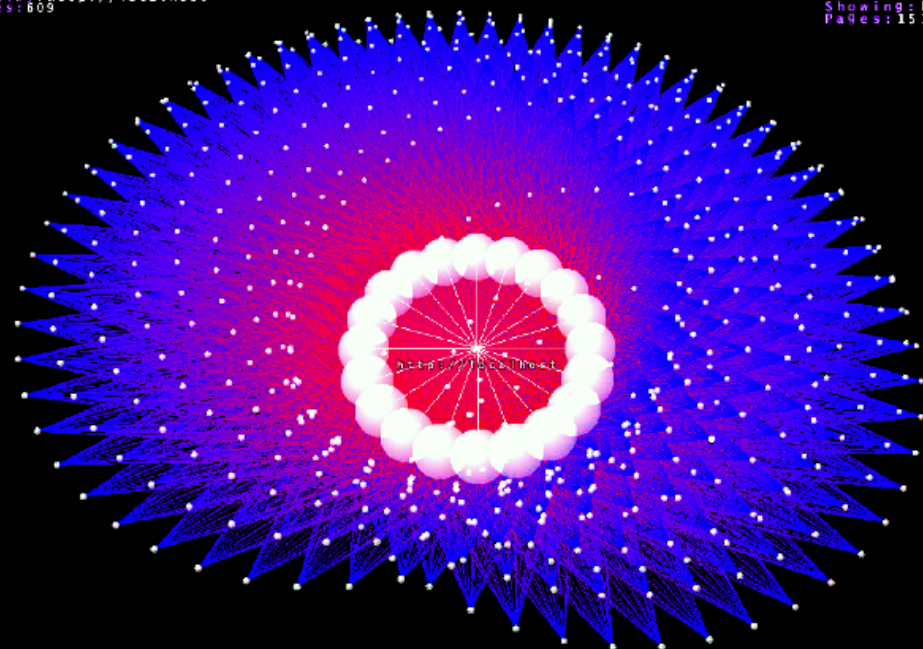
This paper explores software analysis as an artistic practice by examining Tom Betts' *WebTracer2 from 2003*, which is itself an artwork and a browser that makes web structures visible. Through dynamic and static analysis, this work investigates how input travels through a system from visible HTML to invisible machine states and how each transformation generates new forms. By constructing artificial web simulations, manipulating internal file formats, and finally triggering a buffer overflow, the study reveals how software can become both instrument and material through the practice of hacking. The crash emerges as a new generative potential: a point where deterministic logic dissolves and the system becomes reconfigurable. When malformed HTML reshapes control flow and turns *WebTracer2* into an entirely different program, the divide between data and code collapses. The result is a generative act not of images, but of the algorithm itself – an artistic exploration of visibility, instability, and systemic transformation.

KUNST GIBT NICHT DAS SICHTBARE WIEDER, SONDERN MACHT SICHTBAR

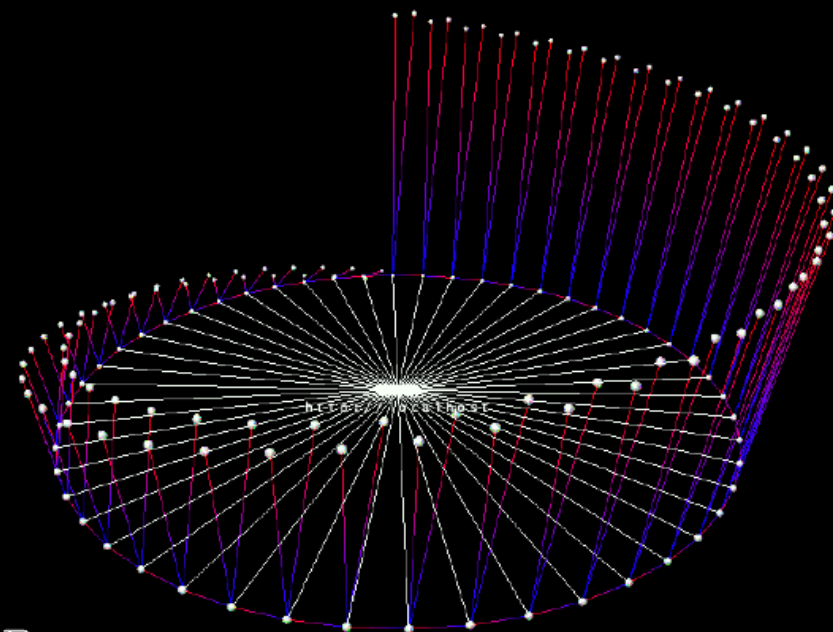
■ KLEE'S DESCRIPTION of abstract painting [1] can equally be applied to net art. The artists who, in the mid-nineties and early two-thousands, began engaging with the emergence of the World Wide Web were above all interested in one aspect: the search for hidden structures behind the obvious and beyond the surface.

general concepts and ideas from the realm of computer hacking.

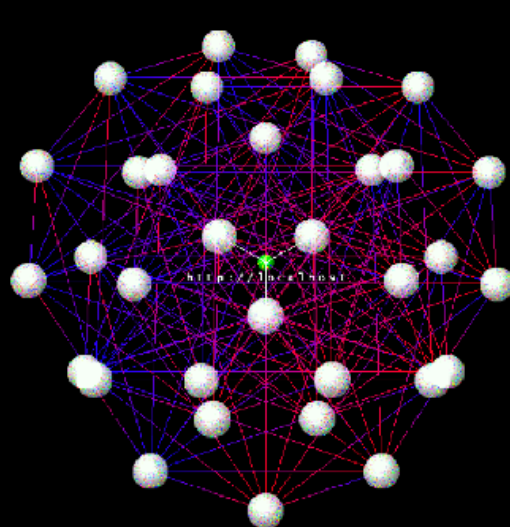
As a case study, I examined *WebTracer2*, a software by Tom Betts (nullpointer) [2]. This software may be regarded within the genre of "net art," and more specifically within the subgenre of "browser art." *WebTracer2* serves as an example for this article because it embodies the very act of making structures visible. It attempts to illustrate the world wide web (WWW) and especially the countless links and interconnections created by hyperlinks, through graphical visualization. Its visual language is built on molecular and cellular metaphors, which was described



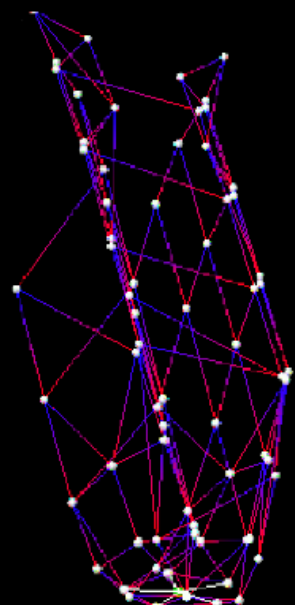
A



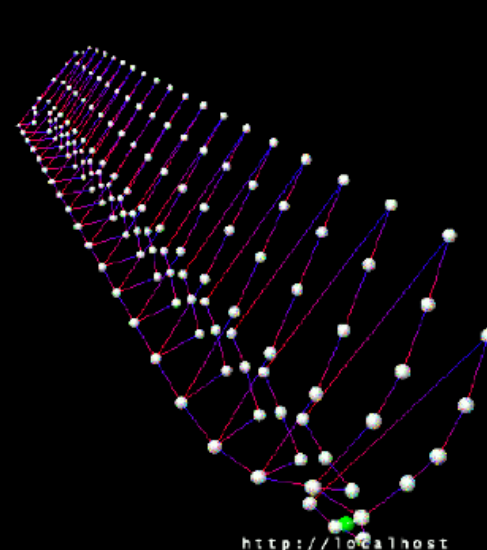
B



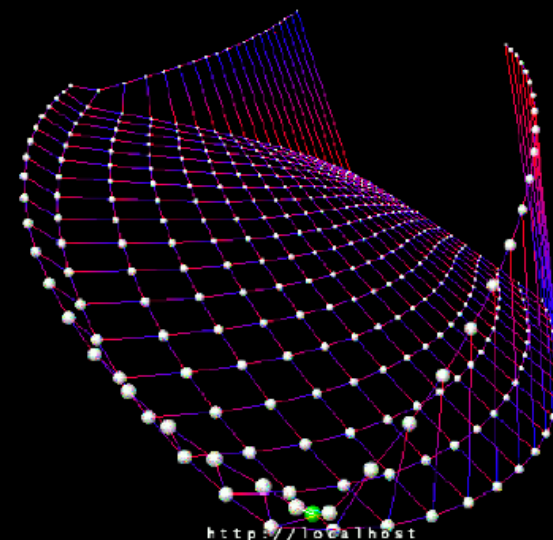
C



D



E

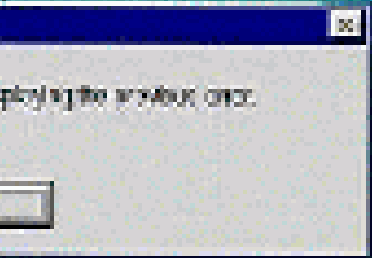


F



```
1 <html>
2 <head><TITLE> %20Location </TITLE></head>
3 <BODY BGCOLOR="#000000" TEXT="#00ff00" LINK="#00ff00" VLINK="#00ff00" ALINK="#ffffff"><CENTER><h4><A HREF="00010/index.html" target="_top">
4
5 </A><blink><B><A HREF="100/index.html" target="_top" onMouseOver="self.status=' =====
6
7 [2]----->
8
9
10
11
12
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14
15
16
17
18
19
20
21
22
23
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25
26
27
28
29
30
31
32
33
34
35
36 [3]
37
38
39
40
41
42 [4]----->
43
44
45
```

<blink> ???





Netomat Reverse Engineering: A failing forward attempt to debug net art

Yannick Westphal

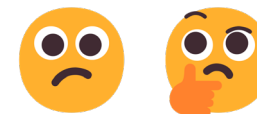
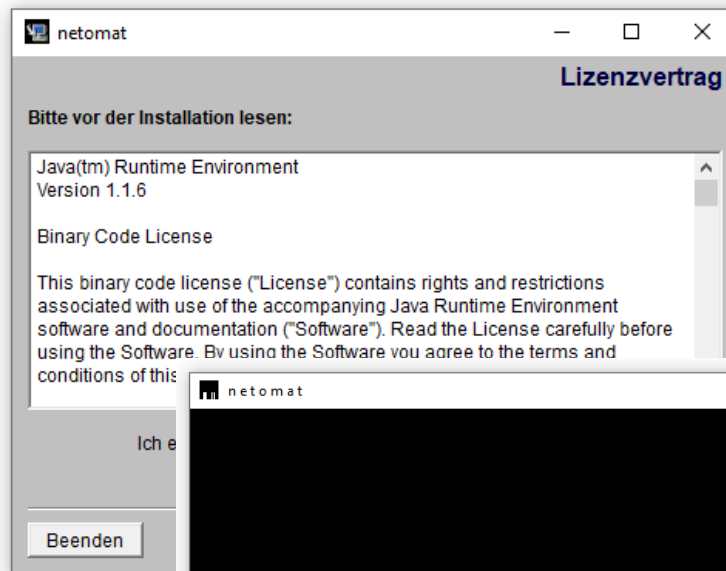
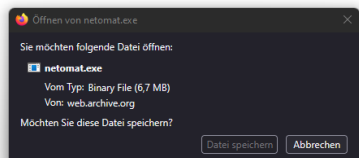
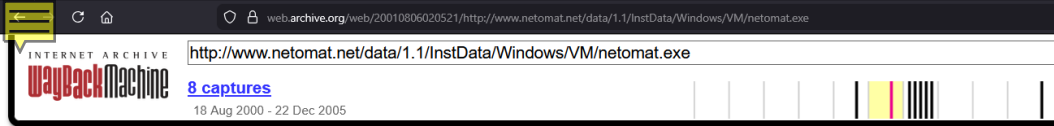
Netomat (1999) by Maciej Wiśniewski — is a net art browser that renders the internet as a continuous, undifferentiated flow of text, images, and sound — a poetic counterproposal to conventional web navigation. Today, the original software no longer runs: its server infrastructure has vanished, its internal communication protocol remains undocumented, and the program fails immediately with a cryptic network error. This article presents a technical investigation of Netomat using reverse engineering methods drawn from software development and IT security. By combining network traffic analysis, Java decompilation, and hands-on debugging, the study reconstructs the program's internal logic and recovers its undocumented Netomatic Markup Language (NML). The process follows a principle of "failing forward" — each error revealing a new piece of information about the system. The result is a partially restored, runnable instance of the original artwork, and a replicable methodology for technically engaging with software-based net art whose execution environment has been lost.

■ THERE are no longer many images or videos on the internet that show the original art browser NETOMAT by Maciej Wiśniewski (1999) in action. Ultimately, the work as net art browser has not survived the constantly changing internet in its original form. Only fragments remain: isolated images and artifacts, and a few articles that surface here and there when you deliberately search archives and art platforms. But this is likely the fate of almost everything that rises and sinks again online, in a continuous flow of data, images, texts, and experiences. This is why it is curiously fitting, given that Netomat was designed to capture precisely this

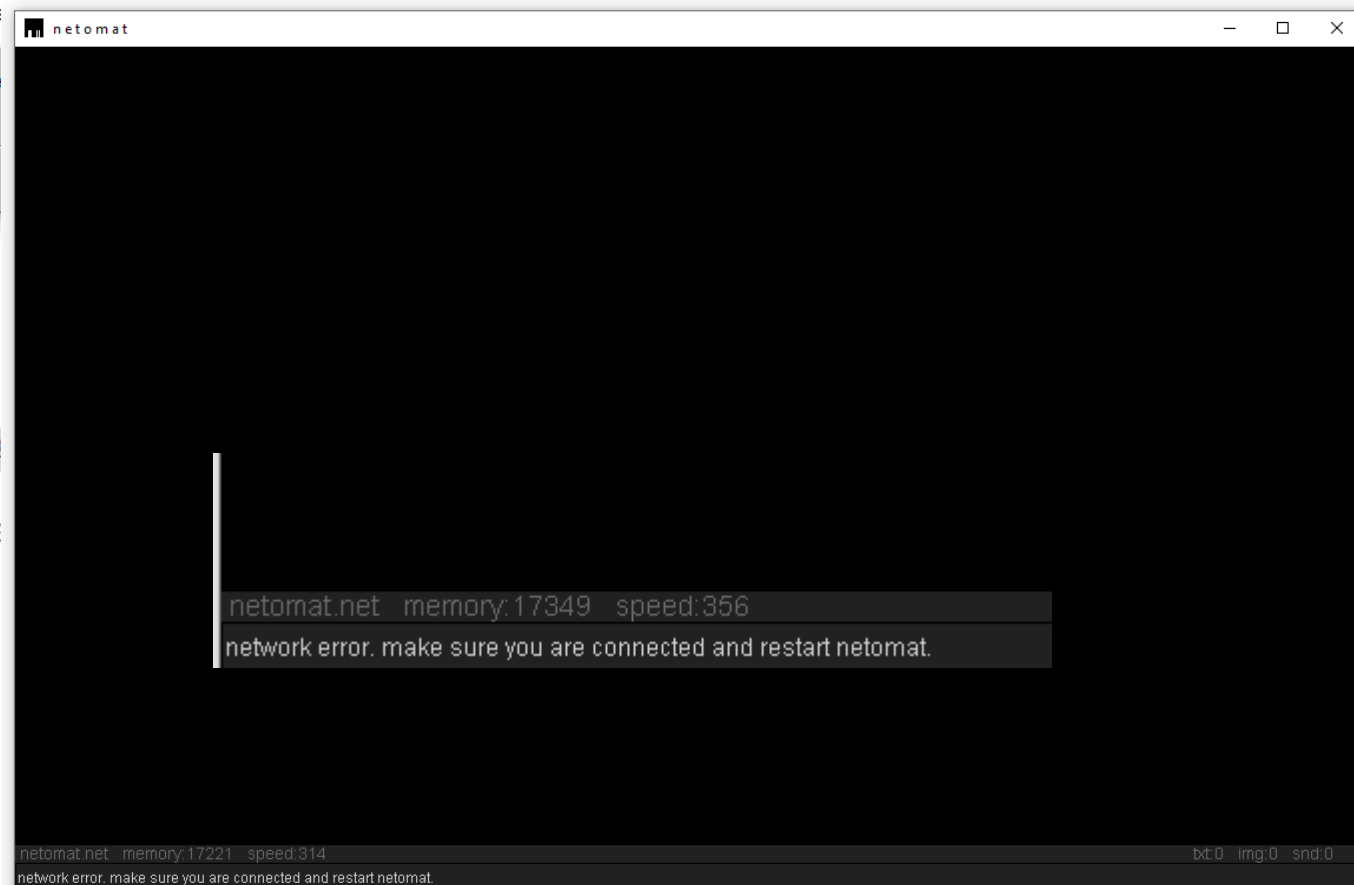
turn of the millennium. But a runnable version of the original program in its first form no longer exists. Even though the work's premise may be more relevant than ever in today's media landscape, literally flooded with "content."

This tension between artistic significance and technical obsolescence poses a challenge: how can a digital artwork be meaningfully studied when its software execution seems lost? When the supposedly repeatable performance of code suddenly produces *errors*?

This article approaches that question by treating



Name	Änderungsdatum	Typ
images	22.03.2026 20:52	Dateiordner
jre	22.03.2026 20:52	Dateiordner
lib	22.03.2026 20:52	Dateiordner
UninstallerData	22.03.2026 20:52	Dateiordner
lax.jar	22.03.2026 20:52	JAR-Datei
LICENSE.TXT	22.03.2026 20:52	Textdokument
netomat.exe	22.03.2026 20:52	Anwendung
netomat.ico	22.03.2026 20:52	Symbol
netomat.jar	22.03.2026 20:52	JAR-Datei
netomat.lax	22.03.2026 20:52	LAX-Datei
netomatics.dtd	22.03.2026 20:52	Document Type D...
properties.netomat	22.03.2026 20:52	NETOMAT-Datei
README.TXT	22.03.2026 20:52	Textdokument
version.netomat	22.03.2026 20:52	NETOMAT-Datei



tcp.stream eq 1						
No.	Time	Source	Destination	Protocol	Length	Info
4	0.7075...	10.0.2.15	75.101.167.104	TCP	66	62596 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
5	0.8135...	75.101.167.104	10.0.2.15	TCP	60	80 → 62596 [SYN, ACK] Seq=0 Ack=1 Win=32768 Len=0 MSS=1460
6	0.8136...	10.0.2.15	75.101.167.104	TCP	54	62596 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
7	0.8141...	10.0.2.15	75.101.167.104	HTTP	198	GET /data/init1.netomat HTTP/1.0
8	0.8621...	75.101.167.104	10.0.2.15	TCP	60	80 → 62596 [ACK] Seq=1 Ack=145 Win=32624 Len=0
9	0.9241...	75.101.167.104	10.0.2.15	HTTP	453	HTTP/1.1 301 Moved Permanently (text/html)
10	0.9241...	75.101.167.104	10.0.2.15	TCP	60	80 → 62596 [FIN, ACK] Seq=400 Ack=145 Win=32624 Len=0
11	0.9242...	10.0.2.15	75.101.167.104	TCP	54	62596 → 80 [ACK] Seq=145 Ack=401 Win=63841 Len=0
12	0.9245...	10.0.2.15	75.101.167.104	TCP	54	62596 → 80 [FIN, ACK] Seq=145 Ack=401 Win=63841 Len=0
13	0.9246...	75.101.167.104	10.0.2.15	TCP	60	80 → 62596 [ACK] Seq=401 Ack=146 Win=32623 Len=0

> Frame 7: 198 bytes on wire (1584 bits), 198 bytes captured (1584 bits) on interface \Device\NPF_{94BA7CF5-A2D3-446F-A8A1-EF3E...}

> Ethernet II, Src: PcsCompu_8c:c9:33 (08:00:27:8c:c9:33), Dst: RealtekU_12:35:00 (52:54:00:12:35:00)

> Internet Protocol Version 4, Src: 10.0.2.15, Dst: 75.101.167.104

> Transmission Control Protocol, Src Port: 62596, Dst Port: 80, Seq: 1, Ack: 1, Len: 144

> Hypertext Transfer Protocol

```

GET /data/init1.netomat HTTP/1.0\r\n
User-Agent: Java1.2.2\r\n
Host: www.netomat.net\r\n
Accept: text/html, image/gif, image/jpeg, *; q=.2, */*; q=.2\r\n
\r\n
[Full request URI: http://www.netomat.net/data/init1.netomat]
[HTTP request 1/1]
[Response in frame: 9]

```

Wireshark · Folge TCP Stream (tcp.stream eq 1) · Ethernet

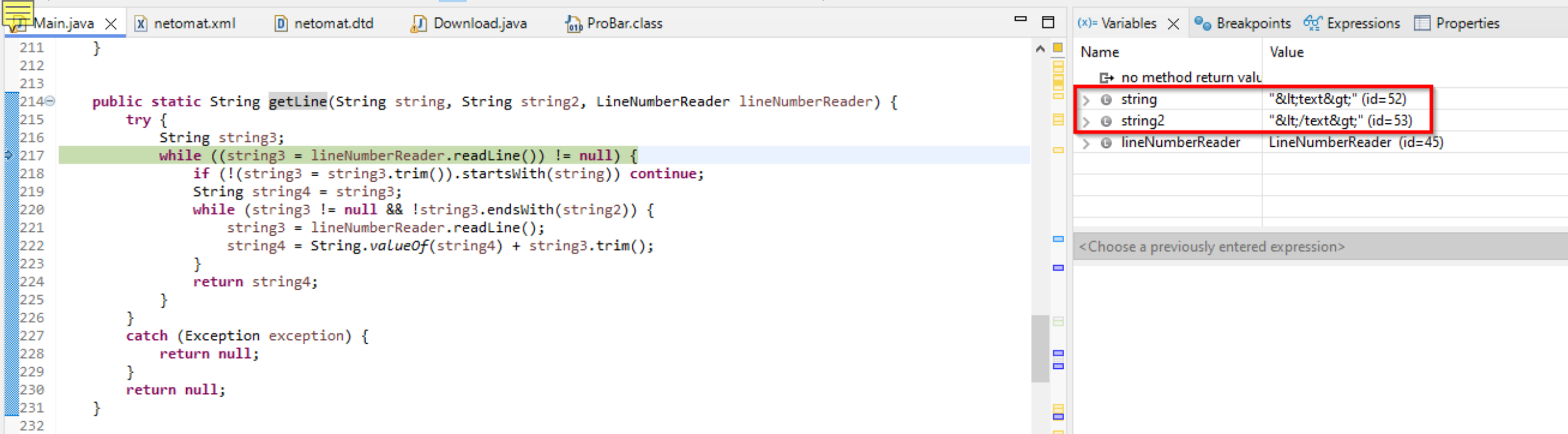
```

GET /data/init1.netomat HTTP/1.0
User-Agent: Java1.2.2
Host: www.netomat.net
Accept: text/html, image/gif, image/jpeg, *; q=.2, */*; q=.2

HTTP/1.1 301 Moved Permanently
Server: nginx/1.18.0 (Ubuntu)
Date: Thu, 07 Nov 2024 12:03:20 GMT
Content-Type: text/html
Content-Length: 178
Connection: close
Location: https://www.netomat.net/data/init1.netomat

<html>
<head><title>301 Moved Permanently</title></head>
<body>
<center><h1>301 Moved Permanently</h1></center>

```



Reverse Engineered Netomatic Markup Language $\leadsto$ NML

```
<service>test=properties.netomat</service>
```

<text>test</text>

```
<image>http://127.0.0.1/data/hackingart.jpg</image>
```

<sound>http://hacking.art/test.mp3</sound>



Knowbotic Reverse: Reconstructing the Server Infrastructure from Disassembling Net Art

Yannick Westphal

Connective Force Attack (2001) — by Knowbotic Research is a networked software artwork designed to coordinate distributed brute-force attacks against a public information system in Hamburg a conceptual intervention into questions of hacking, privacy, and the contested public domain. Today, the original server infrastructure is offline, rendering the software largely non-functional. This article presents a systematic reverse engineering analysis of the client application *cfa.exe*, employing static and dynamic analysis methods drawn from IT security practice: binary inspection, disassembly, network traffic analysis, and debugger-assisted code tracing. By recovering the undocumented communication protocol between client and server a numeric command-response system operating on port 6666 it becomes possible to reconstruct a functional server substitute and restore the artwork's core interface to an operational state. The process is further validated through direct collaboration with the original developer, who provided the surviving server source code. The study demonstrates how reverse engineering methods can serve as both a conservation strategy and a hermeneutic tool for understanding software-based net art whose operational context has been lost.

■ A PARTICULAR IRONY is embedded in *Connective Force Attack*. The work by Knowbotic Research (Yvonne Wilhelm, Alexander Tuchacek, Christian Huebler) was conceived as an act of collective infiltration — a piece of software that would invite internet users to join forces, pool computational power, and crack a password-protected server in Hamburg in order to place their own content on public screens in the city's subway stations. Its entire premises rested on connectivity, on the capacity

servers are gone. The network it once addressed no longer exists in the form it requires. The artwork designed to hack public infrastructure has itself become inaccessible.

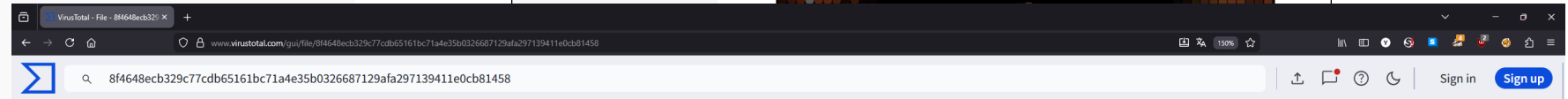
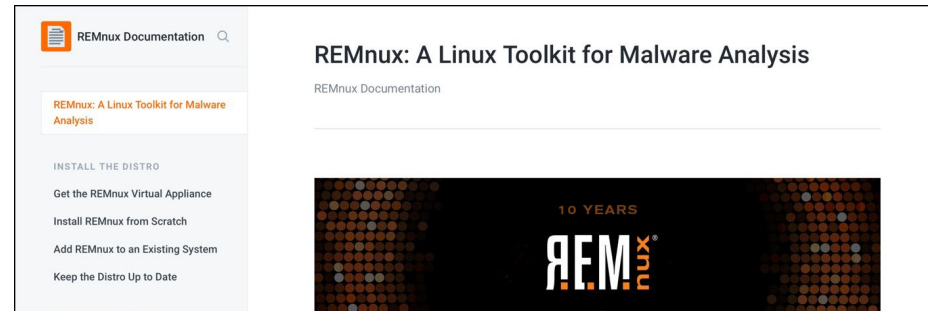
This situation — technically broken, historically significant — raises a question that extends beyond *Connective Force Attack* specifically: what does it mean to study a software-based artwork when its execution environment has collapsed? Art historical methods can describe the work, contextualize it, and



GOBLIN.exe

Dateityp: Anwendung (.exe)

Beschreibung: GOBLIN.exe



0

/ 63

Community Score

✓ No security vendors flagged this file as malicious

Reanalyze Similar More

8f4648ecb329c77cdb65161bc71a4e35b0326687129afa297139411e0cb81458

GOBLIN.exe

Size: 220.92 MB | Last Analysis Date: 4 years ago

88 EXE

peexe overlay direct-cpu-clock-access detect-debug-environment runtime-modules checks-user-input self-delete

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis				Do you want to automate checks?	
Acronis (Static ML)	✓ Undetected	Ad-Aware	✓ Undetected		
AhnLab-V3	✓ Undetected	Alibaba	✓ Undetected		
ALYac	✓ Undetected	Antiy-AVL	✓ Undetected		
Arcabit	✓ Undetected	Avast	✓ Undetected		
Avira (no cloud)	✓ Undetected	Baidu	✓ Undetected		
BitDefender	✓ Undetected	BitDefenderTheta	✓ Undetected		
Bkav Pro	✓ Undetected	ClamAV	✓ Undetected		
CMC	✓ Undetected	Comodo	✓ Undetected		
CrowdStrike Falcon	✓ Undetected	Cybereason	✓ Undetected		

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Aktuelle Bedrohungen

Bedrohungen gefunden. Starten Sie die empfohlenen Aktionen.

Virus:Python/Biennale 31.07.2021 21:37 (Aktiv)	Schwerwiegend
Aktionsoptionen: ☐ Quarantäne ☐ Bedrohung bereinigen ☐ Entfernen ☒ Auf Gerät zulassen Details anzeigen	
Virus:Python/Biennale 31.07.2021 21:37 (Aktiv)	Schwerwiegend
Virus:Python/Biennale 31.07.2021 13:14 (Aktiv)	Schwerwiegend
Virus:Python/Biennale 31.07.2021 13:14 (Aktiv)	Schwerwiegend

Aktionen starten



CONTACT

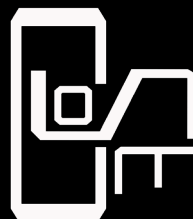
[MAIL\[AT\]HACKING.ART](mailto:MAIL[AT]HACKING.ART)

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- > FREELANCE
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 - VARIOUS ART AND DESIGN COMMISSIONS
 - VARIOUS EXHIBITIONS IN AN ART CONTEXT



CODED SECRETS RESEARCH GROUP

ABOUT ME

> YANNICK WESTPHAL

GERMAN ARTIST AND RESEARCHER CURRENTLY BASED IN BERLIN, WORKING AT THE INTERSECTION OF HACKING, IT SECURITY, AND DIGITAL ART. WITH A DIPLOMA IN MEDIA ARTS (CUM LAUDE), HIS PRACTICE MERGES NET.ART, CYBER AESTHETICS, AND INVESTIGATIVE DIGITAL FORENSICS.

OUTSIDE THE STUDIO, HE SPENT 8+ YEARS IN IT SECURITY AND CONDUCTED 150+ PENETRATION TESTS ACROSS NETWORKS, SYSTEMS, WEB & CLIENT APPS, AND COMPLEX ENTERPRISE ENVIRONMENTS.

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